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Tesla versus BYD: China's Electric Vehicle Contest

Kat, a young private equity investment adviser specializing in the electric vehicle (EV) industry, was preparing a proposal. Her clients were considering making a sizable investment in the Build Your Dreams Company (BYD), the leading player in China's fast-growing EV market. Although virtually unknown in the West, BYD was the largest EV manufacturer in the world, by volume. Offering a range of battery electric vehicles (BEVs) and plug-in hybrid electric vehicles (PHEVs) across multiple price points, BYD operated almost exclusively in the highly subsidized domestic Chinese EV market. BYD was the undisputed leader in that market, but whether it had the technological expertise and capabilities to compete on a global scale with the likes of Tesla, Inc. (Tesla), Volkswagen AG, and General Motors Company (GM) was unclear.

As she considered how to advise her clients, Kat realized that the dynamics and outcome of the competition between BYD and Tesla in China's BEV segment would be good indicators of how the industry would likely develop—and whether BYD could realize its global ambitions. Both manufacturers offered extensive product lines in China. EV pioneer Tesla had entered the Chinese market in 2013 and had faced intense competition from a slew of local players such as BYD; Nio, Inc. (Nio); and Xpeng Motors (Xpeng). Indeed, despite its global brand and charismatic CEO, Elon Musk, Tesla had so far managed to capture a market share of just 8.8% in China.[1] In an earnings call in January 2023, Musk grudgingly acknowledged the threat posed by Tesla's Chinese competitors. "They are the most competitive in the world," he told listeners. "They work the hardest and they work the smartest…probably some company out of China is the most likely to be second to Tesla."[2]

The question for Kat and her clients was not whether BYD would be second, but whether BYD would be first. The outcome of the battle between Tesla and BYD in China, Kat thought, would be a strong indicator of who would prevail in the global EV market in the long run.

The Evolution of the EV

In simplest terms, an EV was defined as a vehicle that used one or more electric motors (rather than a gasoline-powered internal combustion engine) for propulsion. Although EVs had been around since the late 19th century, the internal combustion engine had been the dominant means of propulsion for cars and trucks for well over a century. Early 20th-century passenger car manufacturers preferred gasoline-fueled cars over EVs

[1] "One in Four Cars Sold in China in 2022 Was an EV with BYD Powering Country's Outperformance," Counterpoint Research press release, March 15, 2023, https://www.counterpointresearch.com/china-ev-sales-2022/ (accessed Dec. 5, 2023).

[2] Arjun Kharpal, "Elon Musk Says a Chinese Automaker Will Likely Be Second to Tesla: 'They Work the Smartest,'" CNBC, January 26, 2023. https://www.cnbc.com/2023/01/26/elon-musk-says-a-chinese-automaker-will-likely-be-second-to-tesla.html (accessed Aug. 1, 2023).

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due to highly developed gas station infrastructure, low gasoline prices, and the relative technological maturity of the internal combustion engine.

In the first few decades of the 2000s, EVs had begun to emerge as a way forward due to advances in battery technology and increasing awareness of the environmental impact of fossil fuels. In response to the Paris Climate Agreement,[3] the United States and multiple European governments rolled out incentives aimed at increasing consumer adoption of EVs in the later years of the first decade of the 21st century. By 2022, a total of 14% of all new cars sold in the United States and Europe were electric, up from 9% in 2021 and less than 5% in 2020.[4] EV sales were anticipated to account for more than 35% of all new car sales by 2030,[5] up from just 1% in 2016. In July 2022, the global EV market size was $280 billion, and it was expected to grow to $1 trillion by 2026.[6] (See **Exhibits 1** and **2** for EV sales data.)

Broadly speaking, there were two types of EVs on the market as of 2023:

PHEVs were powered by electric motors and internal combustion engines working in tandem. The electricity for the motor came from a battery pack that could be recharged via a wall plug or by regenerative brakes, which converted a car's kinetic energy into electrical energy. Because drivers could switch between the electric motor and the internal combustion engine, they were not subject to the range limitations of BEVs. The electric motor and battery combination offered greater efficiency and superior gas mileage compared to vehicles with traditional internal combustion engines. The most common application of the PHEV approach in the United States was the Toyota Prius, although almost all major manufacturers had begun to offer hybrid variants of their gasoline-powered models as of 2023.

BEVs were powered exclusively by electric motors, with electricity coming from onboard battery packs, solar panels, or fuel cells. These vehicles had no internal combustion engine, and therefore they produced no greenhouse gas emissions. The challenges associated with BEVs were the cost and size of the necessary batteries and the limited driving range, which generally varied from 150 to 250 miles on a single charge. Acceptance of the pure EV as a long-term replacement for gasoline-powered vehicles was premised on reductions in battery costs and improvements in battery capacity. The most common application of the BEV approach in the United States was the Tesla range of vehicles. Many industry analysts felt that PHEVs were a transitional technology and that the path forward was elimination of the internal combustion engine and a full transition to BEVs.

The most valuable component of the BEV was its battery, typically representing 35% to 40% of the total cost of the vehicle.[7] Crucial raw materials in battery production included lithium, cobalt, nickel, and other rare earth metals, which were primarily sourced from mines in China and Africa. Analysts expected that China would "continue to be the major supplier of battery-grade raw materials" into the next decade.[8] Beyond the battery, EVs relied on intricate software and electronic control systems that fulfilled functions ranging from battery management to motor control, regenerative braking, connectivity, and autonomous driving. The other

[3] The Paris Agreement was a legally binding international treaty on climate change. It was adopted by 196 parties at the UN Climate Change Conference (COP21) in Paris, France, in December 2015.

[4] "Global EV Outlook 2023," International Energy Agency, 2023, https://www.iea.org/reports/global-ev-outlook-2023 (accessed Jun. 25, 2024).

[5] "The Electric Vehicle Outlook Is Stronger in China and Europe Than in the United States," McKinsey & Company, October 19, 2020, https://www.mckinsey.com/featured-insights/sustainable-inclusive-growth/chart-of-the-day/the-electric-vehicle-outlook-is-stronger-in-china-and-europe-than-in-the-united-states (accessed Dec. 5, 2023).

[6] Rob Dillan, "Electric Vehicle (EV) Industry Statistics and Forecasts," EVHYPE, August 30, 2022, https://evhype.com/ev-industry-stats/ (accessed Aug 5, 2023).

[7] "EVs Are Still 45% More Expensive to Make Than Combustion-Engined Cars," InsideEVs, https://insideevs.com/news/444542/evs-45-percent-more-expensive-make-ice/ (accessed May 7, 2024).

[8] "Battery Supply Chain Challenges," Raw Materials Information System, European Commission, https://rmis.jrc.ec.europa.eu/analysis-of-supply-chain-challenges-49b749 (accessed Dec. 5, 2023).

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components of the EV, such as the body, chassis, suspension, brakes, and electric motors, were assembled in highly automated manufacturing facilities like Tesla's Gigafactories.

The establishment of a robust and user-friendly charging infrastructure, such as networks of fast-charging stations with standardized connectors, would play a pivotal role in promoting the widespread adoption of EVs. To maximize the full ecological benefits of EVs, the charging stations would also need to leverage renewable energy sources, such as by integrating solar panels at charging stations. In some cases, EV manufacturers like Tesla had also ventured into residential energy generation and storage solutions, fostering the development of a broader sustainable energy ecosystem. As EV usage increased, many governments were expected to mandate battery recycling, creating a parallel growth industry in battery recycling solutions.[9]

The Chinese EV Market

By 2022, China had already become the largest market for electric cars,[10] with 6.89 million EV sales and 25.6% market penetration in 2022.[11] The Chinese government saw EVs as a significant step toward meeting the requirements of the Paris Climate Agreement and had provided numerous subsidies over the last two decades to promote the manufacture and sale of EVs. At the current growth rate, EV sales were expected to constitute up to 28% of all new automotive sales by the end of 2023,[12] following the end of China's zero-COVID-19 restrictions.

EV technology investment had been an integral part of China's Five-Year Plan, the country's highest-level economic blueprint, since 2001. The government had used a unique "double credits" system for domestic automotive manufacturers. The system set an aggressive corporate average fuel consumption (CAFC) target for all vehicles produced by a given manufacturer. The aim was to incentivize large automotive makers to produce "new energy vehicles" (NEVs), allowing them to earn NEV credits to ensure that their overall sales mix conformed to CAFC guidelines. These NEV credits could then be openly traded, allowing manufacturers of traditional gasoline-powered vehicles to buy credits to meet CAFC standards, and EV manufacturers to sell credits to fund research and development efforts and improve net profit. Additionally, the central government subsidized the construction of charging stations in target cities and actively promoted the use of EVs for public transport.

Between 2009 and 2012, before there was meaningful consumer demand for EVs in China, the government had initiated a project called Ten Cities, Thousand Vehicles. In accordance with the project, the government provided financial support to 10 cities to purchase 1,000 EV buses every year for the public transportation system. In addition, to encourage demand for EVs, drivers in big cities could immediately obtain licenses to drive EVs, while drivers of petroleum-powered cars often needed to spend several thousand dollars to get a license or spend years waiting in a lottery system.[13]

In China, people generally bought cars to commute between their home and office, preferring to make use of the country's network of high-speed trains for long-distance travel. Accordingly, EVs in the Chinese market tended to have a shorter range per charge compared with those in the United States, where long interstate road

[9] Dario Maradin, Ana Malnar, and Ana Kaštelan, "Sustainable and Clean Energy: The Case of Tesla Company," *Journal of Economics, Finance, and Management Studies* 5, no. 12 (December 2022): 3,531–42.

[10] Neil King, "EVs Forecast to Account for Two Thirds of Global Light-Vehicle Sales in 2035," EV Volumes, https://ev-volumes.com/news/ev/evs-forecast-to-account-for-two-thirds-of-global-light-vehicle-sales-in-2035/ (accessed Jun. 25, 2024).

[11] BYD Company Limited Annual Report, 2022, https://www.bydglobal.com/en/InvestorAnnals.html (accessed Dec. 5, 2023).

[12] "One in Four Cars Sold in China in 2022 Was an EV with BYD Powering Country's Outperformance," Counterpoint Research press release, March 15, 2023, https://www.counterpointresearch.com/insights/china-ev-sales-2022/ (accessed Jun. 25, 2024).

[13] Wenbo Li, Muyi Yang, and Suwin Sandu, "Electric Vehicles in China: A Review of Current Policies," *Energy & Environment* 29, no. 8 (December 2018): 1,512–24.

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trips were the norm and range anxiety (i.e., concern that a battery's charge wouldn't be sufficient to travel the necessary distances) continued to be a major factor in adoption.

The Chinese EV market was highly fragmented and competitive, with more than 94 brands and 300 models ranging in price from $5,000 to over $90,000.[14] In 2022, BYD was the market leader, with 27% market share.[15] Its closest competitor was SAIC-GM-Wuling Automobile (SGMW), a joint venture involving US-based GM, China-based SAIC Motor Corp., Ltd. (SAIC), and Liuzhou Wuling Automobile Industry Co. (Wuling), which had 8.9% market share.[16] Tesla, which entered the market in 2017, ranked a close third with 8.8% market share.[17] Not only was BYD the market-share leader in China but it had also edged out Tesla to become the number-one global EV manufacturer by volume, selling 1.8 million BEVs and PHEVs in 2022.[18] Nonetheless, Tesla remained the global market-share leader for BEVs by a large margin, with a 17% share of all BEVs sold. (For 2022 market shares and 2023 sales in China, see **Exhibits 3** and **4**.)

Beyond those major players, there was a wide range of EV start-ups in China offering lower-end models at competitive price points, including Nio, Xpeng, Neta, AITO, IM Motors, Zeekr, Aiways Automobiles Company, Ltd, and Livan Automotive.[19] Government subsidies were expected to end in 2023, and the market seemed poised to enter a period of consolidation, as many of the smaller players had yet to become profitable. BYD, however, was a notable exception. Its combined EV and handset components business achieved a healthy 3.9% net profit margin in 2022. This was a marked improvement from its 1.3% margin in 2019—an increase that was largely attributable to its thriving EV business.[20]

Build Your Dreams

From rechargeable batteries to EVs

BYD was founded in 1995 in Shenzhen, China, by Wang Chuanfu and his cousin, Lu Xiangyang. Wang was born into a farming family in rural Anhui Province. Trained as a chemist, he earned a bachelor's degree in metallurgical engineering. He also obtained a master's degree at the Beijing General Research Institute for Nonferrous Metals and went on to work on battery technologies for the Chinese government. In 1995, Wang teamed up with Lu to establish BYD, a rechargeable battery manufacturer based in Shenzhen. The company was immensely successful, becoming the first Chinese supplier of lithium-ion batteries for Motorola, Inc., in 2000. In 2002, BYD was publicly listed on the Hong Kong Stock Exchange, and in 2004, it expanded into the automobile industry through the acquisition of the struggling state-owned Qinchuan Automobile Company.

BYD developed its first BEV prototype, the F3e, in 2005. But China's battery-charging infrastructure in major cities was almost nonexistent, and the car was never put into production. In 2008, BYD launched the F3DM, a PHEV with a combination gasoline engine and battery that allowed it to work around the limited number of charging stations. That same year, BYD was boosted by a $230 million investment from Berkshire Hathaway Inc., which took a 10% stake in the company.[21] At the time, BYD was primarily focused on

[14] Oleksiy Soroka and Rico Luman, "Car Market Outlook 2023: Seeing Is Believing," ING Think, January 17, 2023, https://think.ing.com/articles/car-market-outlook-2023-seeing-is-believing/ (accessed Dec. 5, 2023).
[15] https://www.bydglobal.com/en/InvestorAnnals.html.
[16] https://www.counterpointresearch.com/insights/china-ev-sales-2022/.
[17] https://www.counterpointresearch.com/insights/china-ev-sales-2022/.
[18] Marcus Lu, "Global EV Production: BYD Surpasses Tesla," Visual Capitalist, April 20, 2023, https://www.visualcapitalist.com/global-ev-production-byd-surpasses-tesla/ (accessed Sept. 15, 2023).
[19] Roland Irle, "Global EV Sales for 2022," EV Volumes, February 6, 2023, https://ev-volumes.com/news/ev/global-ev-sales-for-2022/#:~:text=Non%2Dchargeable%20variants%20were%20phased,in%20all%20BEVs%20sold%20worldwide (accessed May 7, 2024).
[20] https://www.bydglobal.com/en/InvestorAnnals.html.
[21] Alex Crippin, "Warren Buffett Invests in Chinese Company Developing 'Green' Cars," CNBC, September 27, 2008, https://www.cnbc.com/2008/09/27/warren-buffett-invests-in-chinese-company-developing-green-cars.html (accessed Sept. 10, 2023).

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commercial sales, such as electric taxis and buses for local governments that qualified for the centrally funded Ten Cities, Thousand Vehicles program. BYD leveraged this early experience and switched to selling passenger vehicles in 2019. More significantly, the company had developed deep expertise in battery technology and could manufacture its own lithium iron phosphate (LFP) batteries.[22] This resulted in a significant cost advantage compared to Tesla, which had to outsource production of its higher-cost cobalt aluminum oxide (NCA) batteries. Notably, Tesla eventually switched to lower-cost LFP batteries sourced from BYD for its Model 3 vehicle.[23]

By 2022, BYD had become the world's largest manufacturer of EVs, achieving $46.4 billion in revenue from its automotive business.[24] However, its product portfolio remained largely focused on lower-priced vehicles (i.e., between $15,000 and $25,000), resulting in gross margins between 13% to 17% between 2017 and 2022.[25] While these were comparable to traditional automotive companies such as GM and Ford Motor Company (Ford), they paled in comparison with the likes of Tesla.[26] This issue would become more acute when the Chinese government ceased subsidizing the EV industry in 2023.

Proliferation of products

BYD had a far more complicated product portfolio than Tesla. It offered traditional gasoline vehicles, PHEVs, and BEVs through 15 different product lines and multiple sub-brands. In contrast to Tesla, which had entered the middle of the market with the Model 3, BYD was now looking to launch higher-end models priced beyond $35,000. In 2023, BYD had boldly announced the availability of two luxury cars priced at more than $150,000: the U8 and U9, under the luxury sub-brand Yangwang.[27]

Since 2020, BYD had focused on its Dynasty series and Ocean series, which had contributed the most to its sales. The company named models after Chinese imperial dynasties to boost brand awareness and familiarity among Chinese consumers. The Qin, Han, Yuan, and Tang lines all had models with both BEV and PHEV options. Among the three, the Qin fell at the lower end of the price range, with the Qin model starting as low as $12,000, while the Han and Tang series had models starting at $30,000 and $35,000 respectively. The Yuan was the cheapest BEV in the Dynasty series, with starting prices as low as $16,000. The Ocean series, targeted for overseas markets, consisted of the Dolphin, Seal, Destroyer, and Frigate. The Destroyer 07 and Frigate 07 were both PHEVs with prices starting at $22,000 and $29,000, respectively.[28]

Despite its efforts to offer more premium products, BYD continued to be largely perceived as a "value brand." Consumers still associated it with rechargeable batteries and its early mass-market EV products. Even with its current product portfolio, the majority of BYD's models were still priced under $25,000—far less than Tesla's, which sold for an average of $50,000. In addition, BYD continued to use a distributor network to expand its coverage and was not able to replicate the uniformly premium direct-sales model that Tesla offered.

[22] Matt Blois, "Lithium Iron Phosphate Comes to America," *Chemical & Engineering News*, January 29, 2023, https://cen.acs.org/energy/energy-storage-/Lithium-iron-phosphate-comes-to-America/101/i4 (accessed May 5, 2023).

[23] Michael Wayland, "Tesla Will Change the Type of Battery Cells It Uses in All Its Standard-Range Cars," CNBC, October 20, 2021, https://www.cnbc.com/2021/10/20/tesla-switching-to-lfp-batteries-in-all-standard-range-cars.html (accessed May 5, 2024).

[24] https://www.bydglobal.com/en/InvestorAnnals.html.

[25] https://www.bydglobal.com/en/InvestorAnnals.html.

[26] Tesla, in comparison, had gross margins of nearly 26%. See Tesla, Inc., annual report, 2022.

[27] "BYD Presents Several New Models at Auto Shanghai 2023," BYD press release, April 18, 2023, https://www.byd.com/us/news-list/BYD-Presents-Several-New-Models-at-Auto-Shanghai-2023.html (accessed Jul. 5, 2024).

[28] Dev Shah, "BYD Cars & SUVs: A Detailed Look at the EV Maker's Global Portfolio," March 19, 2024, https://www.team-bhp.com/news/byd-cars-suvs-detailed-look-ev-makers-global-portfolio (accessed May 5, 2024).

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Tesla

Starting with BEVs

Tesla was founded in 2003 by American entrepreneurs Martin Eberhard and Marc Tarpenning, who served as Tesla's first CEO and CFO, respectively. Unlike traditional automotive companies, Tesla never offered an internal combustion engine or PHEV product but rather jumped straight into BEVs. Elon Musk provided much of Tesla's initial funding and assumed the role of board chair. In 2006, Musk published the first articulation of Tesla's long-term mission: "To help expedite the move from a mine-and-burn hydrocarbon economy towards a solar electric economy" by offering "a wide range of models, including affordably priced family cars." Musk became CEO in 2008, and the company went public in 2010, the first automaker to do so in the United States since Ford in 1956.[29]

South Africa–born Musk emigrated to Canada as a teenager and obtained bachelor's degrees in economics and physics from the University of Pennsylvania. A natural entrepreneur, in 1995, he started online business directory service Zip2 Corp. (Zip2). Zip2 was acquired by Compaq Computer Corporation in 1999, and Musk rolled his capital into X.com, an online financial services company. X.com merged with Confinity Inc. to form PayPal Holdings, Inc., and was purchased by eBay Inc. in 2002. Musk had since been involved in multiple start-ups, including Space Exploration Technologies Corp. (SpaceX), SolarCity Corporation, and X, formerly known as Twitter.

As the world dealt with the myriad difficulties of the COVID-19 pandemic in 2020, Tesla finally turned its earnings positive, garnering $721 million in net income on $31.5 billion in revenue. The company grew rapidly, achieving revenue of $53.8 billion in 2021 and $81.4 billion in 2022. Moreover, it enjoyed industry-leading gross margins of up to 26% in 2022. The revenue surge came from burgeoning consumer interest in electric cars coupled with the availability of the relatively more affordable Model 3 and Model Y, and Tesla was poised for growth.[30] Factors contributing to the company's impressive margins included its high average selling price, which still exceeded $50,000 in 2021; its direct-sales model, which avoided profit sharing with third-party distributors; superior economies of scale, as the Model 3 and Model Y were essentially built on the same platform; and its state-of-the-art, highly automated Gigafactories.

More than just cars

Tesla had started out with its Roadster EV in 2008, followed by its Model S EV in 2012. Both vehicles were well received but aimed at the luxury market, often costing over $100,000. The Roadster, intended to be a showpiece product, was discontinued, while the Model S continued to be upgraded and improved. The Model X SUV, launched in 2015, was perhaps the most iconic of the Tesla vehicles, with dramatic gull-wing-style rear passenger doors and an expansive integrated windscreen-roof. The Model X sat at the top of the Tesla lineup, costing $100,000 to $120,000, depending on options.

The launch of the Model 3 was a milestone for Tesla, marking a move into the junior executive sedan segment occupied by the likes of the gasoline-powered BMW 3 series, Mercedes C class, and Audi A4. The Model 3 was a smaller version of the Model S, having a shorter and narrower cabin with a more accessible target price of $35,000 for the base model. The Model 3 was Tesla's biggest seller, appealing to customers who wanted the Tesla experience without the six-figure price tag. Tesla's Model Y, along with the SUV version of the Model 3, targeted the compact crossover/SUV segment; both were launched in 2020. The most basic version of the Model Y cost about $47,000—low for Tesla, but still much higher than Chinese-made

[29] Q.ai, "Tesla: A History of Innovation (and Headaches)," *Forbes*, September 29, 2022, https://www.forbes.com/sites/qai/2022/09/29/tesla-a-history-of-innovation-and-headaches/?sh=1d3d047a1873 (accessed Aug. 15, 2023).

[30] Tesla, Inc., annual report, 2022.

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equivalents.[31] Tesla had implemented a highly successful product strategy of establishing its price anchor and luxury brand image with the Model S and Model X, then generating revenue from the lower-priced Model Y and Model 3, which constituted over 90% of sales.[32]

In addition to offering cutting-edge vehicles, Tesla was also a pioneer in autonomous vehicle technology. The Autopilot system, a standard feature in every Tesla, helped the vehicle stay in its lane and adjust its speed to match traffic conditions. An Enhanced Autopilot option allowed the car to autonomously park and change lanes on highways. The most advanced autonomous option was the Full Self-Driving (FSD) feature. When fully developed, FSD would not only be able to park, change lanes, and adjust speed, but also stop and start a car according to traffic lights, stop signs, and traffic conditions. The technology was still being fine-tuned, and Tesla offered the caveat that it was intended for use with a "fully attentive driver" prepared to take over at any moment. Tesla saw self-driving technology as the future of the automobile, and Musk, in a 2023 earnings call, said he believed that FSD technology, once perfected, would be a primary profit driver for the company.[33]

Challenges in the Chinese market

Tesla opened its first retail outlet in China in 2013. As of February 2023, its sales had risen 31.6% year-over-year, according to data from China's automotive association, but its market share had declined from 15% in 2020 to 10% in 2022 and to 9% in February 2023.[34] (For Tesla models and prices in China, see **Exhibit 5.**)

Tesla had a strong brand in China, though it ranked only third in terms of market share. However, its generous profit margin allowed more room for price reductions: 26% gross margins were a distant dream for China's EV manufacturers, many of which still struggled to achieve profitability. To continually enhance its cost advantage, Tesla had made a huge investment in China over the past three years. In 2019, Tesla built a Gigafactory in China with annual production capability of 1.1 million cars, leveraging the benefit of lower labor costs and a strong business ecosystem.[35] In 2022, Tesla began ordering LFP batteries from BYD.[36] As the most commonly used battery technology in China, LFPs offered a significant cost advantage over the NCA batteries traditionally used by Tesla.

Still, the EV giant faced many challenges in the Chinese market. Its flagship autonomous driving system struggled with congestion in urban areas and lack of network connection in rural areas. Data privacy and national security were also concerns: the eight cameras used by Tesla's Autopilot and the built-in dashboard camera could monitor a vehicle's surroundings and record footage of anything close by. Tesla moved all data generated by its cars in China to local storage to address these data security concerns.[37] Moving forward, geopolitical tensions between China and the United States would also represent an unavoidable challenge. Tesla's plans to build another factory in Shanghai to produce its Megapack energy storage system were met with criticism in Washington, which had asked multinational companies to reduce their reliance on China-based supply chains. On the Chinese side, Tesla was the only automaker with a factory in China that didn't have a

[31] Tesla website, https://www.tesla.com (accessed Dec. 5, 2023).

[32] Christiaan Hetzner, "Elon Musk Is Cutting Tesla Prices in the U.S. Again after Poor Q3 Sales, Leaving Bulls with 'No Way to Sugarcoat This,'" *Fortune*, October 6, 2023, https://fortune.com/2023/10/06/elon-musk-tesla-price-cuts-q3-sales/ (accessed Jun. 25, 2024).

[33] Peter McGuthrie, "Elon Musk's Long-Term Bet on Tesla Full Self-Driving," CleanTechnica, May 7, 2023, https://cleantechnica.com/2023/05/07/elon-musks-long-term-bet-on-tesla-full-self-driving/ (accessed Sept. 8, 2023).

[34] Mohit Oberoi, "Tesla's China Sales Rise but It Continues to Lose Market Share to BYD," Business2Community, March 3, 2023, https://www.business2community.com/stock-news/tesla-china-sales-rise-but-it-continues-to-lose-market-share-to-byd-02646846 (accessed Aug. 23, 2023).

[35] Fred Lambert, "Tesla Announces Production of 2 Millionth EV at Giga Shanghai," Electrek, September 7, 2023, https://electrek.co/2023/09/07/tesla-announces-production-2-millionth-ev-giga-shanghai/ (accessed Jun. 25, 2024).

[36] Daniel Bleakley, "Tesla's Switch to BYD Batteries Is Achieving Faster Charging Times," The Driven, May 22, 2023, https://thedriven.io/2023/05/22/teslas-switch-to-byd-batteries-is-achieving-faster-charging-times/ (accessed Jun. 25, 2024).

[37] Michelle Toh, "Tesla Sets Up Data Center in China amid Spying Concerns," CNN Business, May 26, 2021, https://www.cnn.com/2021/05/26/business/tesla-china-data-center-intl-hnk/index.html (accessed Jun. 25, 2024).

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joint venture with a state-owned company, and continued support from the Chinese government remained questionable.[38]

SAIC-GM-Wuling Automobile

Formed in 2002, SGMW had captured the second-largest share of the EV market in China by focusing on small, affordable commercial and consumer vehicles sold under the Wuling and Baojun brands. Wuling had existed since 1982 and was best known for its commercial microvans and its Hongguang Mini EV, a three-door hatchback that was the next-best-selling EV after the BYD and Tesla vehicles. The Hongguang Mini EV would be classified in Western markets as a micro-car. A BEV, it was equipped with a minuscule 40-horsepower electric motor and had a limited range of 170 miles, making it ideal for one or two people to make short trips. Unlike the Tesla Model 3 or the BYD Han, the Hongguang Mini EV was not intended to be a full-featured BEV or to be used for interstate driving. In 2020, its starting price was $4,162; it topped out at $5,607 for a fully featured model, making it China's cheapest electric car.[39]

Wuling's other products were more traditional gasoline-powered automobiles. The Baojun brand had been established in 2010 with the aim of selling to consumers in third- and fourth-tier Chinese cities, where it competed with local brands like Chery and Geely. Baojun did not have a significant PHEV or BEV offering.

Recent Competitive Moves

Tesla's competitive actions in China were largely focused on establishing a stable, cost-effective manufacturing base centered on its state-of-the-art, hyperefficient Gigafactory in Shanghai. Construction had begun in 2018, and Tesla had started ramping up production and selling China-made cars in 2019.[40] By 2020, Tesla's $6.66 billion in China sales accounted for just over a fifth of the $31.54 billion total market. The company's Model 3 was the bestselling electric car in the country in 2020, according to China's Passenger Car Association.[41] News agency Reuters calculated that about half the 936,000 vehicles Tesla delivered in 2021 came from its Gigafactory in Shanghai, exceeding the number built in its Fremont, California, facility.[42] In addition, Tesla cut the price of its Model 3 and Model Y by 9% to increase demand.

During the same period, BYD had adopted a different strategy, becoming more focused on expanding its product range—including entering the luxury EV market previously dominated by Tesla.[43] BYD launched several new models on both ends of the price spectrum, such as the higher-end Han and Frigate 07 models and the more affordable Seal and Destroyer models. (For a list of BYD models and prices, see **Exhibit 6**.) Most notably, BYD launched its Han series of vehicles, which were intended to directly compete with the Model 3. With prices between $29,000 and $41,000,[44] the Han was an all-new executive sedan with a rapid-charging LFP

[38] Rebecca Feng, "Tesla to Build New Megapack Battery Factory in Shanghai," *Wall Street Journal*, April 9, 2023, https://www.wsj.com/articles/tesla-to-build-new-china-plant-for-energy-storage-battery-aa8a76db (accessed Jul. 3, 2024).

[39] Yilei Sun and Brenda Goh, "GM Energizes China Line-Up with Electric Micro Car," Reuters, September 25, 2020, https://www.reuters.com/article/us-gm-china-electric-focus-idUSKCN26G0H9/ (accessed Dec. 5, 2023).

[40] Dana Hull and Chunying Zhang, "Elon Musk Set Up His Shanghai Gigafactory in Record Time," Bloomberg, October 23, 2019, https://www.bloomberg.com/news/articles/2019-10-23/elon-musk-opened-tesla-s-shanghai-gigafactory-in-just-168-days (accessed Jun. 25, 2024).

[41] Evelyn Cheng, "Tesla's Sales More Than Doubled in 2020," CNBC, February 8, 2021, https://www.cnbc.com/2021/02/09/teslas-china-sales-more-than-doubled-in-2020.html (accessed Sept. 9, 2023).

[42] Huileng Tan, "Half of Tesla's Global Deliveries in 2021 Came from China, as the EV Maker Moved Its Export Hub to Shanghai from Fremont," *Business Insider*, January 12, 2022, https://www.businessinsider.com/half-of-teslas-2021-deliveries-from-china-shanghai-export-hub-2022-1 (accessed Sept. 1, 2023).

[43] Arjun Kharpal, "Musk Once Laughed Off BYD as a Threat. Now the Chinese Giant Has Taken Tesla's EV Crown—Here's How," CNBC, January 4, 2024, https://www.cnbc.com/2024/01/05/how-byd-grew-from-a-phone-battery-maker-to-ev-giant-taking-on-tesla.html (accessed Jun. 25, 2024).

[44] Scooter Doll, "BYD Fuels EV Price War in China, Cutting MSRPs of Newly Refreshed Tang and Han Models," Electrek, February 28, 2024, https://electrek.co/2024/02/28/byd-fuels-ev-price-war-china-cutting-msrps-refreshed-tang-han-models/ (accessed Jun. 25, 2024).

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battery, a 375-mile range, and all-wheel drive. These new products helped BYD build a complete offering under its "two legs" strategy (i.e., offering both BEVs and PHEVs) and helped establish the brand's credibility in the premium market. As a result, BYD's EV sales, including hybrid vehicles, more than doubled in 2022.[45] According to the China Passenger Car Association, Tesla's sales rose by 50% over the same period.[46]

Tesla continued to cut prices on its Models Y and 3 by 11% and 6%, respectively, reducing their price premium relative to BYD and other local players.[47] In comparison, BYD launched its first luxury EV SUV, the U8, as well as its first electric sportscar, the U9, under its premium sub-brand Yangwang, at prices north of $150,000. While sales of the U8 and U9 were expected to be small, the cars nevertheless showcased BYD's technological prowess and global ambitions. In response, BYD also reduced prices by up to 5%,[48] and it launched the Seagull, which at approximately $10,000 was one of the least expensive BEVs in China.

Tesla released Q1 financials in April 2023, reporting a decline in global profits from $3.3 billion to $2.5 billion over the same period in 2022.[49] On May 1, Tesla marginally increased its prices by $250, signaling an end to the recent slew of price cuts. CEO Musk signaled on an earnings call that Tesla would be targeting larger volumes of sales versus higher margins.[50] Despite the increases, Tesla's vehicles remained less expensive than they had been at the start of the year.[51]

Looking Ahead

How long could BYD maintain its lead over Tesla in the Chinese market? How sustainable was the company's strategy of rapid new vehicle launches? GM and Chrysler Corporation's experiences in the 1980s showed that a proliferation of models and sub-brands could lead to mediocre products, poor differentiation, and poor economies of scale. Would BYD meet the same fate? Similarly, the limitations of Tesla's hyperfocused approach were also apparent: with only two mainstream offerings (i.e., Models 3 and Y), Tesla might not be able to cater to the disparate needs of the Chinese market. Finally, where did SGMW fit into the BEV battle? Was it content to dominate the micro-van and micro-car markets, or would it eventually leverage GM's technological capabilities to compete more broadly with Tesla and BYD?

In the long run, who would succeed? BYD had made a strong showing in a recent German automotive show and was looking to expand into Europe.[52] It was clear that the Chinese BEV market would be a good indicator of the outcome of global EV competition. Kat pondered BYD's and Tesla's likely next steps as she prepared the presentation for her clients.

[45] Wenyi Zhang, "Total Vehicle Sales Volume of BYD Group from FY 2019 to FY 2022, by Type," Statista, January 3, 2024, https://www.statista.com/statistics/1306233/total-vehicle-sales-byd-by-type/ (accessed Jun. 25, 2024).

[46] Jack Ewing, "Tesla Sales in China Slump as Competition Intensifies," *New York Times*, January 5, 2023, https://www.nytimes.com/2023/01/05/business/tesla-china-sales-electric-vehicles.html (accessed Aug. 28, 2023).

[47] Evelyn Cheng, "Tesla Cut EV Prices in China More Than BYD Did for Its Flagship Han Sedan This Year, Study Finds," CNBC, December 20, 2023, https://www.cnbc.com/2023/12/20/tesla-cut-china-ev-prices-more-than-byd-did-for-its-flagship-han-sedan.html (accessed Jun. 25, 2024).

[48] https://www.cnbc.com/2023/12/20/tesla-cut-china-ev-prices-more-than-byd-did-for-its-flagship-han-sedan.html.

[49] "Tesla Releases First Quarter 2023 Financial Results," Tesla, Inc., press release, April 2023, https://ir.tesla.com/press-release/tesla-releases-first-quarter-2023-financial-results (accessed Jun. 25, 2024).

[50] Lora Kolokdny, "Tesla Net Income Drops More Than 20% from Last Year," CNBC, April 20, 2023, https://www.cnbc.com/2023/04/19/tesla-tsla-earnings-q1-2023.html (accessed Jun. 25, 2024).

[51] Arjun Kharpal, "Tesla Hikes Prices in U.S., China and Other Markets after Multiple Cuts," CNBC, May 2, 2023, https://www.cnbc.com/2023/05/02/tesla-raises-car-prices-in-us-china-and-other-markets-after-cuts.html (accessed Sept. 1, 2023).

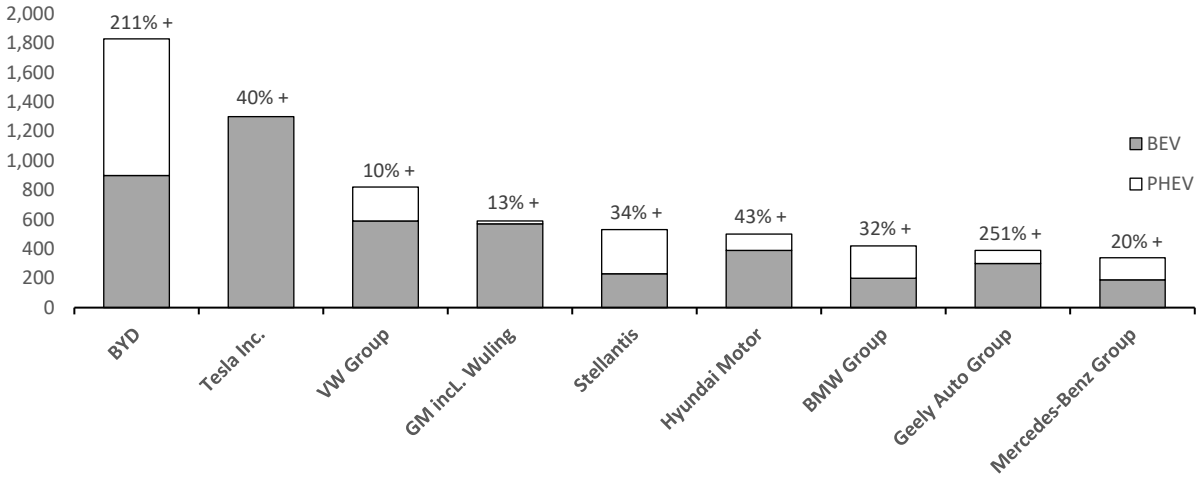
[52] Melissa Eddy, "Chinese Cars Star at Munich Auto Show, Underscoring German Economic Woes," *New York Times,* September 5, 2023, https://www.nytimes.com/2023/09/05/world/europe/chinese-automakers-iaa-mobility-german-auto-show.html (accessed Jul. 5, 2024).

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Exhibit 1

Tesla versus BYD: China's Electric Vehicle Contest

Global Electric Vehicle Sales by Brand (2022) (in thousands)



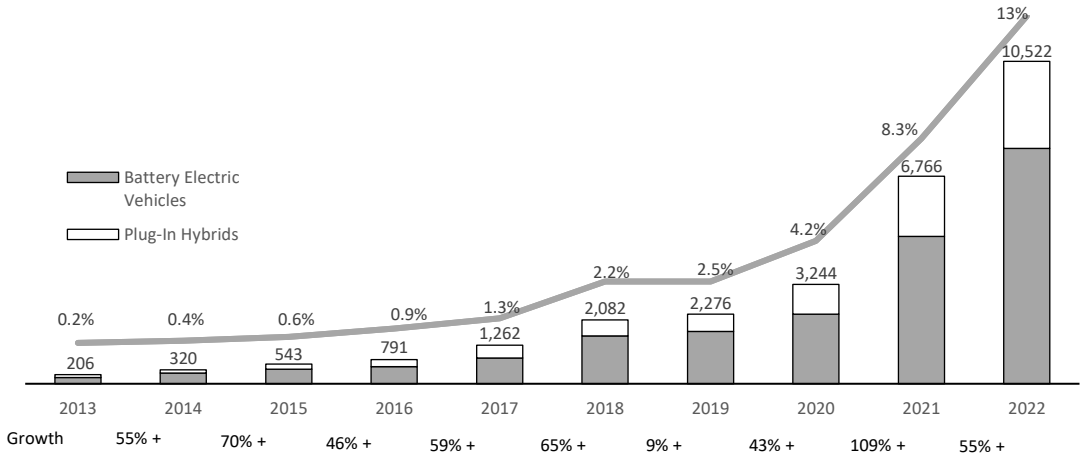
BEV = battery electric vehicle; PHEV = plug-in electric vehicle.

Source: Created by authors based on Neil King, "EVs Forecast to Account for Two Thirds of Global Light-Vehicle Sales in 2035," EV Volumes, https://ev-volumes.com/news/ev/evs-forecast-to-account-for-two-thirds-of-global-light-vehicle-sales-in-2035/ (accessed Jun. 25, 2024).

Exhibit 2

Tesla versus BYD: China's Electric Vehicle Contest

Global BEV and PHEV Sales (in thousands)



BEV = battery electric vehicle; PHEV = plug-in electric vehicle.

Source: Created by authors based on Neil King, "EVs Forecast to Account for Two Thirds of Global Light-Vehicle Sales in 2035," EV Volumes, https://ev-volumes.com/news/ev/evs-forecast-to-account-for-two-thirds-of-global-light-vehicle-sales-in-2035/ (accessed Jun. 25, 2024).

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Exhibit 3

Tesla versus BYD: China's Electric Vehicle Contest

Electric Vehicle Market Share in China, 2022

Company	Market Share
BYD	29.7%
SAIC-GM-Wuling	8.9%
Tesla	8.8%
Geely Auto	5.2%
Changan Automobile	4.5%

Data source: "One in Four Cars Sold in China in 2022 Was an EV with BYD Powering Country's Outperformance," Counterpoint Research press release, March 15, 2023, https://www.counterpointresearch.com/insights/china-ev-sales-2022/ (accessed Jun. 25, 2024).

Exhibit 4

Tesla versus BYD: China's Electric Vehicle Contest

Top-Selling Electric Vehicle Model Sales in China, 2023

Make	Model	Sales		
		BEV	PHEV	Total
BYD	Song	25,053	153,192	178,215
Tesla	Model Y	121,407	0	121,407
BYD	Qin Plus	32,560	86,682	119,242
BYD	Yuan Plus	106,014	0	106,014
BYD	Dolphin	100,009	0	100,009
Wuling	Honguang Mini EV	87,928	0	87,928
Geely Auto	Aion S	67,879	0	67,879
Tesla	Model 3	55,978	0	55,978
BYD	Han	26,609	26,034	52,643
Geely Auto	Aion Y	52,120	0	52,120

BEV = battery electric vehicle; PHEV = plug-in electric vehicle.

Data source: "Top 20 Best Selling Electric Vehicles in China—January–April 2023," graph in José Pontes, "35% Plugin Vehicle Market Share in China! (China EV Sales Report)," CleanTechnica, May 2023, https://cleantechnica.com/2023/05/31/35-plugin-vehicle-market-share-in-china-china-ev-sales-report/ (accessed Dec. 5, 2023).

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Exhibit 5

Tesla versus BYD: China's Electric Vehicle Contest

Tesla Prices in China

Model	Base Price (October 2022) (converted to US dollars)
Model 3	$36,800
Model 3 Performance	$48,400
Model Y	$40,000
Model Y Long Range	$49,500
Model Y Performance	$55,000
Model S	$109,300
Model S Plaid	$139,800

Source: Phate Zhang, "Data Tracker: Tesla China Price Changes," CNEV Post, April 21, 2024, https://cnevpost.com/data-tracker/tesla-china-price-changes/ (accessed Jun. 25, 2024).

Exhibit 6

Tesla versus BYD: China's Electric Vehicle Contest

BYD Product and Price

	Model	2023 Average Base Price (converted to US dollars)	Technology
Dynasty	Qin	$12,000	BEV/PHEV
	Han	$30,000	BEV/PHEV
	Tang	$35,000	BEV/PHEV
	Yuan	$16,000	BEV
	Song Plus	$23,000	PHEV
Ocean	Seagull	$10,000	BEV
	Dolphin	$14,000	BEV
	Seal	$26,000	BEV
	Destroyer 07	$22,000	PHEV
	Frigate 07	$29,000	PHEV

BEV = battery electric vehicle; PHEV = plug-in electric vehicle.

Source: Dev Shah, "BYD Cars & SUVs: A Detailed Look at the EV Maker's Global Portfolio," Team-BHP, March 18, 2024, https://www.team-bhp.com/news/byd-cars-suvs-detailed-look-ev-makers-global-portfolio (accessed Jun. 25, 2024).

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